

NATIONAL INSTITUTE OF TECHNOLOGY KARNATAKA, SURATHKAL
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
Blockchain (CS740), END-SEM EXAM, 2nd Semester-M.Tech

Note:

1. Answer all questions.
2. Write all parts of one question in one place.
3. To the point with proper diagram (if any) answers are expected.

Total: 40 marks

Duration: 3:00 Hr.

Date: April 18, 2023

Q No.	Question	Part marks
1	Write Short Note: a) Crash Fault b) Bitcoin Scripts c) MD4 and MD5 d) PoET	4 X 2 = 8
2	a. Explain Birthday Attack with an example. b. How Birthday Attack is related to Hash Collision? c. What is/are the major key design architectural changes from permissionless to permissioned blockchain? d. "Smart Contract execution should always needs to be deterministic" - True or False- Justify with reason.	2.5+2+1.5+2=8
3	a. Why State machine replication is important for permissioned blockchain? b. Explain all steps for State machine replication. c. If you apply state machine replication in public blockchain then what advantages you will get or difficulties you will face? d. Why distributed consensus? Explain distributed consensus in	2.5+2+2+2.5=9

	detail with its requirements.	
4	a. Why do we require $3f+1$ replicas to ensure safety in an asynchronous system when there are f faulty nodes? b. When view change is required? Explain in detail about View Change mechanism. c. Explain a Use case, where permissioned blockchain will be a suitable candidate with reason. d. Explain Byzantine General Problem and its solution.	$2+3+1.5$ $+3.5=10$
5	a. Explain the analysis of Hashcash PoW. b. How Hashcash extended in Bitcoin PoW?	$2+3=5$

-: Good Luck :-